Human Practice

HP

We expect our project to be **people-oriented**, especially to those who may be overwhelmed by despair. As our name intends to convey, we hope our project could dispel darkness and shed light on the dreaded desease that haunts a great number of people;

From our topic selection to all subsequent designs and event arrangements, we have always tried our best to uphold this original aspiration:

Problem identification

From cases around us to specific data, we have realized that cancer is a global health challenge that is becoming increasingly prevalent. Various types of research aiming to conquer cancer have been carried out vigorously. But there is still work to be done. We hope to utilize the knowledge of synthetic biology and the iGEM platform to contribute to this great cause that concerns all of humanity. When selecting a specific type of cancer as subject, we hope to find the "unmet need" in an area that affects a wide range of people but at the moment has , in a certain degree, failed by current treatments that cannot adequately respond to patients' needs. We aim to develop a new treatment to send the light of hope to these patients who may be lost in the thick fog posed by treatment uncertainty and other challenges.

End users analysis

Our project provides an innovative therapy for SCLC, so it is essential to understand the physiological and pathological knowledge of the subject, such as the causes and onset characteristics of SCLC.

However, **the main focus of our project is serving the SCLC patients rather than just treating deseease**. Therefore, we hope that the therapy we develop is not just a treatment, but also truly takes into account the actual needs of the patients. Thus, we first conduct a detailed analysis of the tipical patients' journey through consulting documentary materials and conducting interviews and surveys *(see the iHP section plan)*:

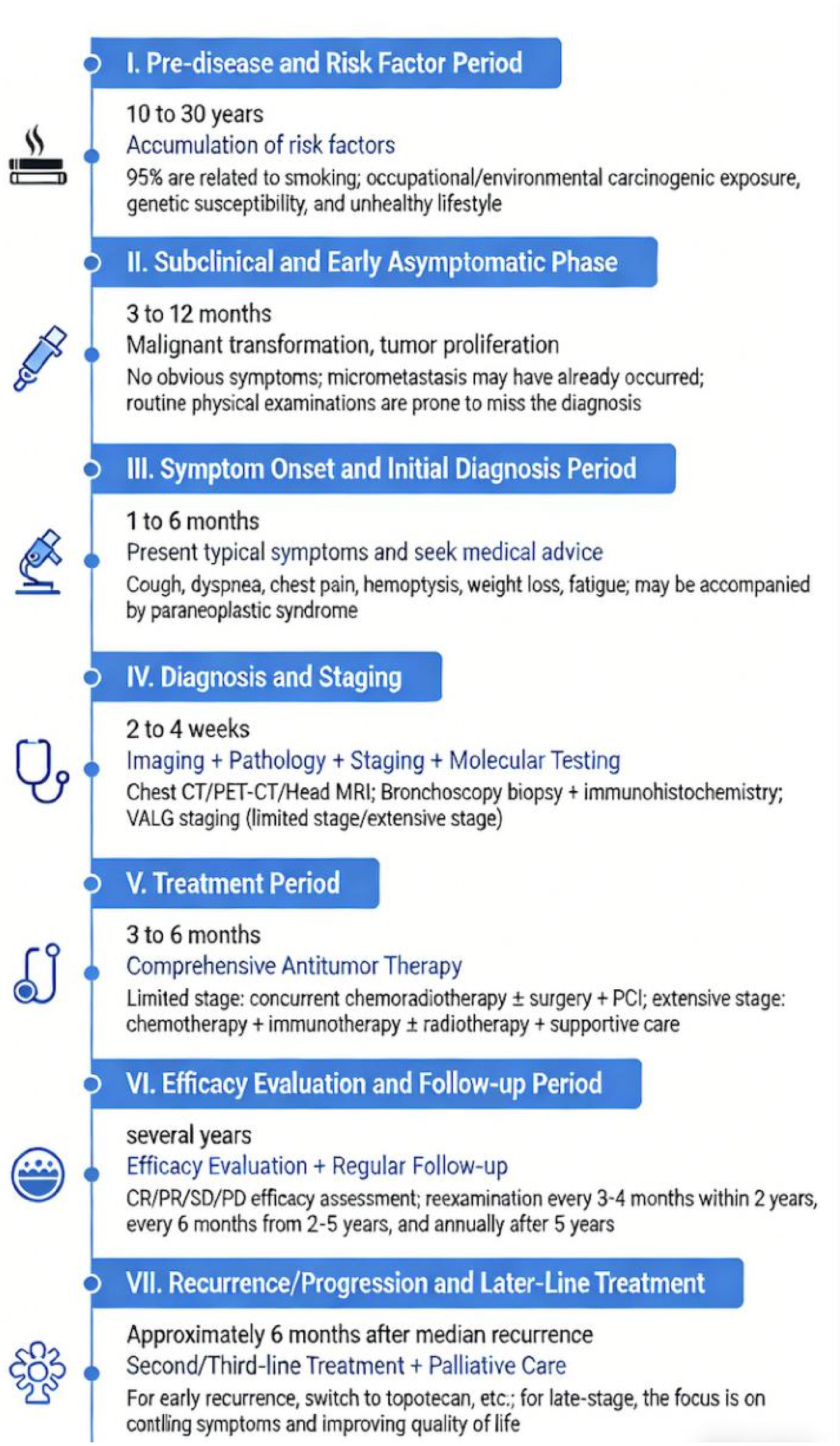

Figure 1 Typical SCLC patient's journey （note:Image generation assisted by AI ）

Currently, the mainstream treatment for SCLC still relies on chemotherapy, which often yields mediocre efficacy. Meanwhile, some major chemotherapy drugs such as topotecan have obvious side effects like hematotoxicity, bringing negative experiences to patients. Therefore, we hope to explore a new drug that is milder and has better drug efficacy targeting. Eventually, we identified a short peptide that induces aggregation, which induces cancer cell death through a brand-new mechanism, significantly reducing side effects while also showing great potential in overcoming drug resistance. At the same time, in the face of another challenge of SCLC, its high recurrence rate, we decided to adopt the administration mode of engineered bacteria colonization to ensure long-term treatment.

To ensure safety, we specifically modified the chassis organism to make it more adaptable to the lung environment, and introduced components such as suicide switches in the design, eliminating the possibility of its escape into the environment or health tissue and ensuring its colonization on tumor sites is alse within containment.

iHP

We plan to further iterate and refine the design after completing the conceptualization of the project prototype. To ensure that the design can address real-world needs, we will conduct in-depth analysis of stakeholders related to the project, carefully understand their needs and suggestions they could provide for our project. Through continuous validation and learning, we can truly confirm that our design is "responsible and good for the world".

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Education & Communication

In the face of cancer, a dreaded disease that many people see as almost a death sentence, we need to gain power from knowledge to help us fignt off fear. By understanding it, we learn how to prevent it in advance and how to better cooperate during treatment. We hope to reduce people's fear of cancer, deal with it rationally, and empower people to protect themselves and help others more effectively through the empowerment of knowledge.

Meanwhile, we also hope to convey the charm of synthetic biology to everyone by disseminating the knowledge and ideas of synthetic biology, taking project design as an example to demonstrate its application potential. We aim to guide students to pay more attention to and explore related fields, and also enable the public to have a more understanding and receptive attitude towards synthetic biology products that may increasingly emerge in the future.

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Collaboration & Partnership

We believe that through cooperation, we can promote our own projects, assist others, and expand the influence of other activities. Therefore, we have decided to collaborate with other iGEM teams to obtain more objective evaluations of our own projects and jointly implement some educational plans, etc.

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Entrepreneurship

Through surveys of patient groups, we have also learned that some new therapies with more effectiveness have poor accessibility for some patients due to cost and trial reasons, making advances in treatment having a had time reaching the mojority of poeple in need. In response to this dilemma, we hope that our products can truly be "patient-friendly" in terms of cost and accessibility. Therefore, we have developed a product implementation plan and intend to engage in dialogues with experts in fields such as medical insurance and pharmaceutical companies during the later stages of the project to discuss how to reduce costs while ensuring quality, and how to map out the project to make the therapy potentially covered by medical insurance.

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